FINAL-YEAR PROJECT

NoteNest

Everything you study, in one place — and it works offline.

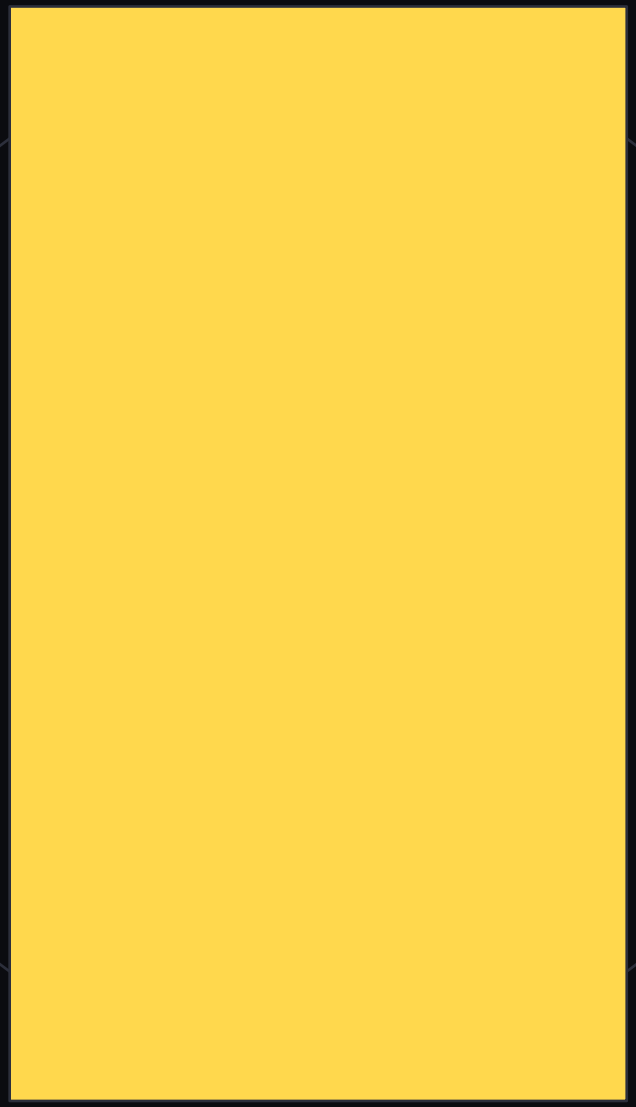
Md. Sarowar Hossain

Student ID .....

Department .....

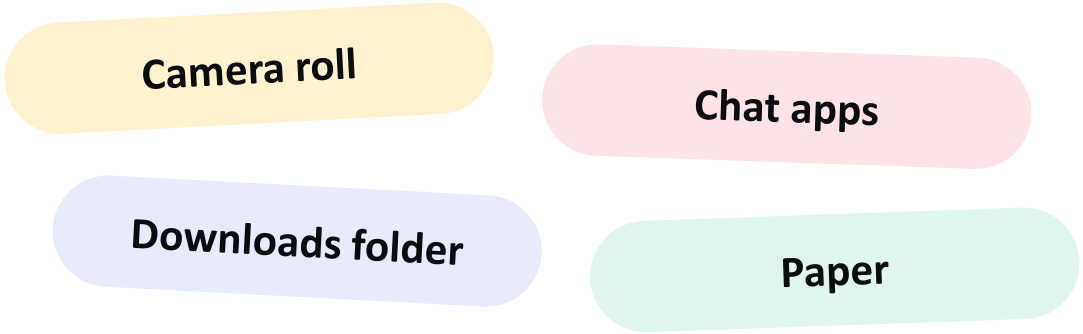
Supervisor .....

Submission date .....



Exam week should be for studying.

Handwritten notes, typed notes, slides as PDFs, whiteboard photos, a recording — all for one lecture.



[statistic — to be cited]

None of these four places knows the items belong to the same lecture.

So exam week is spent searching instead of studying.

One notebook, even offline.

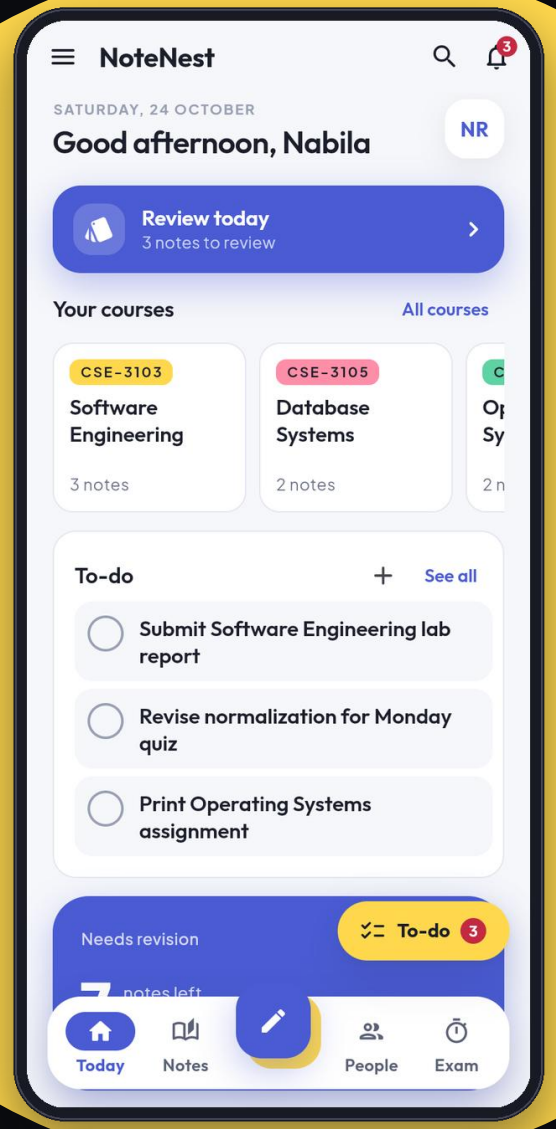
Semester → Course → Note

Every note belongs to a course, every course to a semester. No connection needed.

WHAT MAKES IT DIFFERENT

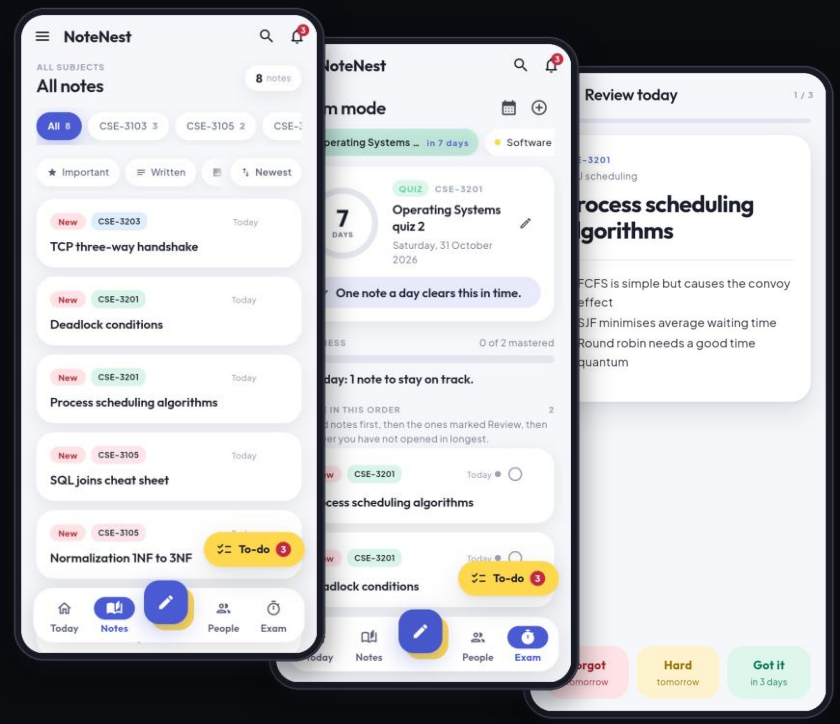
01 Organised by the academic calendar, not by folders

02 The device is the source of truth, not a server



What it does.

01	Notes filed under courses
02	Photos, PDFs and voice on one note
03	Search that reaches inside PDF text
04	Spaced revision
05	Exam countdown and to-dos
06	Cover-page generator
07	Word and PDF conversion
08	Share a note with a classmate
09	Group chat and classes
10	Backup to your own Google Drive
11	PIN and fingerprint lock
12	Bangla and English, light and dark





Spaced revision: 1, 3, 7, 14, then 30 days

Five screens, start to finish.

1

Sign in

2

Degree setup

3

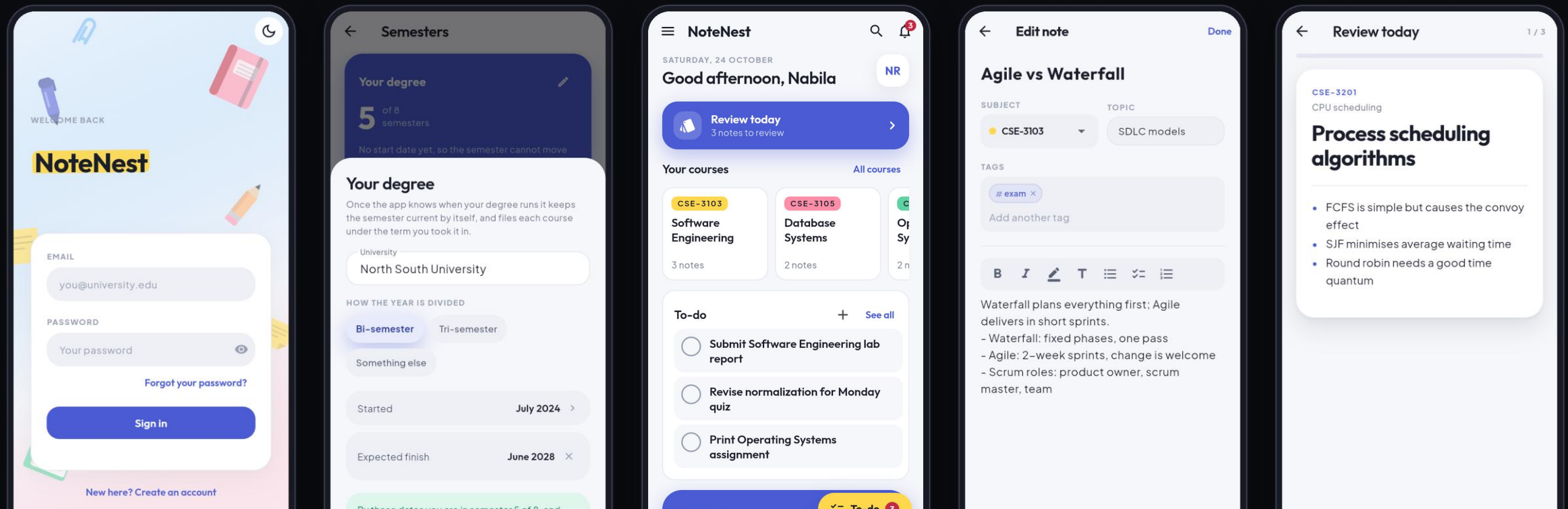
Today

4

Note editor, with attach bar

5

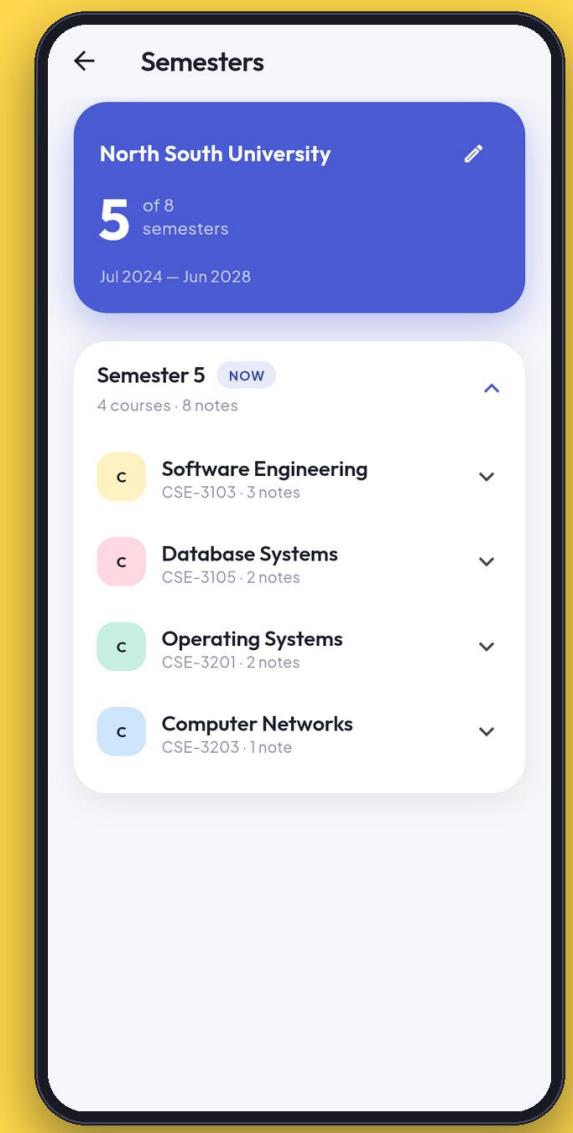
Review



Who it is for.

PRIMARY	University students who take notes on a phone and study for scheduled exams.
SECONDARY	Teachers, through the class feature.
WHY THESE USERS	The app is built around semesters and exams, which only matter to people on an academic calendar.

[number of students — to be cited from UGC / university statistics]



How it is built.

ON THE STUDENT'S PHONE

Flutter and Dart
One codebase

SQLite through sqflite

12 tables · schema version 14
Foreign keys on
The source of truth

accounts, chat, classes

files, backup

Firebase · free Spark plan

Authentication
Firestore
Cloud Messaging
Optional: the app opens without it

The student's own Google Drive

Attachments and backup
drive.file scope only

There is no server of our own, so the Firestore security rules carry all of the authorisation.

Compared with what exists.

PRODUCT	DOES WELL	WHY IT DOES NOT SOLVE THIS
Google Keep	Fast, simple notes with colour and labels	Organised by colour and label, with no idea of a course or an exam.
Evernote	Powerful search, built for professional work	Made for professional work; the free plan has limits.
Notion	Flexible pages and databases	Built online-first; working offline is limited.
Google Classroom	Teachers post work, students hand it in	A teacher's tool, not a notebook for the student's own notes.
Paper notebook	No battery, no signal	Cannot be searched, and holds no PDF or recording.

1 **Organised by the academic calendar**

2 **Offline is the normal state, not a failure**

Free, and cheap to run.

01 Free to use: no advertising, no subscription.

02 Firebase's free plan carries accounts, chat and classes.

03 Attachments go to each student's own Google Drive, so the storage cost does not grow as users are added.

THE REAL LIMITS OF THE FREE (SPARK) PLAN

No Cloud Storage.

No Cloud Functions; the push-notification function in the project needs the paid plan.

Daily Firestore quotas.

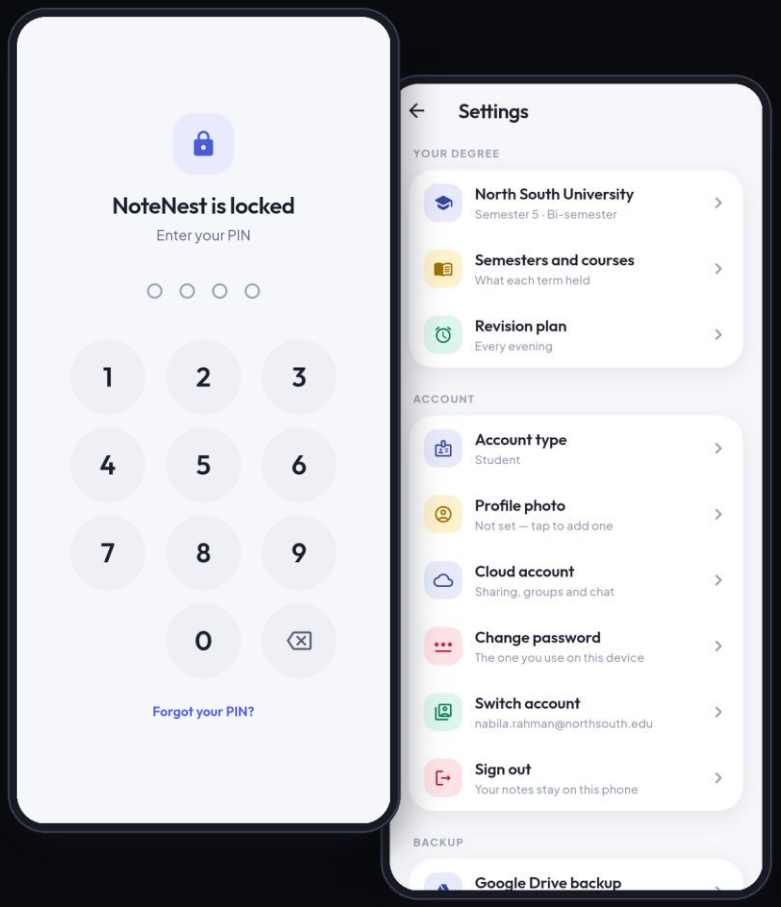
IF IT EVER OUTGREW THE FREE PLAN

The next step would be the paid Blaze plan.

Estimated monthly cost .....

Security and privacy.

PASSWORDS	Never stored. Only a salted PBKDF2-HMAC-SHA256 hash, 120,000 rounds.
ACCOUNTS APART	Every row of study data carries its owner, and every query filters on it. Two accounts on one phone cannot see each other.
GOOGLE DRIVE	drive.file: the app sees only files it created. The rest of the Drive is invisible.
AUDIT	My own code, audited before release: nine findings fixed, including a user-directory leak and the weak password hash.
PLAY REQUIREMENTS	A privacy policy is published and account deletion is built in.



Where it stands, and what is next.

50,152	283	12 · v14	9
lines of Dart in 141 files	automated tests, all passing	tables · schema version	audit findings fixed

STATUS

Built, signed with an upload key, version 1.0.2. Not yet published on the Play Store.

NOT DONE YET

Notes do not sync between phones.

Android only.

No field testing with real users.

The 34 Firestore rules tests are run by hand, not automatically.

ROADMAP

Sync notes between devices

Run the rules tests automatically

Publish on Google Play

An iOS build

Measure performance and accessibility

NoteNest keeps a student's work in one place, organised by course, and it works whether or not there is a signal.

Thank you

Email .....

Phone .....

GitHub .....

Designed and built by Md. Sarowar Hossain

